

```
-- =====
```

```
-- Layer \: SaaS Business Layer (REVISED & SECURED) - PART \
```

```
-- =====
```

```
CREATE EXTENSION IF NOT EXISTS pgcrypto;
```

```
-- \. tenants
```

```
CREATE TABLE tenants (
```

```
tenant_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```
tenant_code VARCHAR(\ . .) NOT NULL,
```

```
tenant_name VARCHAR(Ȳ . .) NOT NULL,
```

```
legal_name VARCHAR(Ȳ . .),
```

```
tenant_type SMALLINT NOT NULL DEFAULT \,
```

```
slug VARCHAR(\ . .) NOT NULL,
```

```
primary_domain_enabled BOOLEAN NOT NULL DEFAULT FALSE,
```

```
domain_status SMALLINT NOT NULL DEFAULT \,
```

```
status SMALLINT NOT NULL DEFAULT \,
```

```
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
```

```
created_by UUID,
```

```
updated_at TIMESTAMPTZ,
```

```
updated_by UUID,
```

```
deleted_at TIMESTAMPTZ,
```

```
deleted_by UUID,
```

```
row_version BIGINT NOT NULL DEFAULT \
```

```
);
```

```
CREATE UNIQUE INDEX uq_tenants_code ON tenants(tenant_code) WHERE deleted_at IS NULL;
```

```
CREATE UNIQUE INDEX uq_tenants_slug ON tenants(slug) WHERE deleted_at IS NULL;
```

```
-- ۲. tenant_wallets
```

```
CREATE TABLE tenant_wallets (
```

```
  wallet_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```
  tenant_id UUID NOT NULL REFERENCES tenants(tenant_id) ON DELETE RESTRICT,
```

```
  balance NUMERIC(۲۰,۴) NOT NULL DEFAULT ۰, -- ارتقا به نوع داده دقیق مالی
```

```
  status SMALLINT NOT NULL DEFAULT ۱,
```

```
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
```

```
  created_by UUID,
```

```
  updated_at TIMESTAMPTZ,
```

```
  updated_by UUID,
```

```
  deleted_at TIMESTAMPTZ,
```

```
  deleted_by UUID,
```

```
  row_version BIGINT NOT NULL DEFAULT ۱
```

```
);
```

```
CREATE UNIQUE INDEX uq_tenant_wallet ON tenant_wallets(tenant_id) WHERE deleted_at IS NULL;
```

```
-- ۳. plans
```

```
CREATE TABLE plans (
```

```
  plan_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```
  code VARCHAR(۵۰) NOT NULL,
```

```
  name VARCHAR(۲۰۰) NOT NULL,
```

```
  status SMALLINT NOT NULL DEFAULT ۱,
```

```
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
```

```
  created_by UUID,
```

```

updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT \
);

CREATE UNIQUE INDEX uq_plans_code ON plans(code) WHERE deleted_at IS NULL;

```

-- ξ. plan_versions

```

CREATE TABLE plan_versions (
    plan_version_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    plan_id UUID NOT NULL REFERENCES plans(plan_id) ON DELETE RESTRICT,
    version_number INT NOT NULL,
    status SMALLINT NOT NULL DEFAULT \,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    created_by UUID,
    updated_at TIMESTAMPTZ,
    updated_by UUID,
    deleted_at TIMESTAMPTZ,
    deleted_by UUID,
    row_version BIGINT NOT NULL DEFAULT \
);

CREATE UNIQUE INDEX uq_plan_version ON plan_versions(plan_id, version_number) WHERE
deleted_at IS NULL;

```

-- ο. plan_prices

```

CREATE TABLE plan_prices (

```

```

plan_price_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

plan_version_id UUID NOT NULL REFERENCES plan_versions(plan_version_id) ON DELETE
RESTRICT,

amount NUMERIC(10,2) NOT NULL,

billing_period_days INT NOT NULL,

status SMALLINT NOT NULL DEFAULT 1,

created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

created_by UUID,

updated_at TIMESTAMPTZ,

updated_by UUID,

deleted_at TIMESTAMPTZ,

deleted_by UUID,

row_version BIGINT NOT NULL DEFAULT 1

);

CREATE INDEX idx_plan_prices_version ON plan_prices(plan_version_id) WHERE deleted_at IS
NULL;

```

-- 7. plan_modules

```

CREATE TABLE plan_modules (

plan_module_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

plan_version_id UUID NOT NULL REFERENCES plan_versions(plan_version_id) ON DELETE
RESTRICT,

code VARCHAR(50) NOT NULL,

name VARCHAR(255) NOT NULL,

status SMALLINT NOT NULL DEFAULT 1,

created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

created_by UUID,

```

```

updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT \
);

```

```

CREATE INDEX idx_plan_modules_version ON plan_modules(plan_version_id) WHERE
deleted_at IS NULL;

```

```

-- V. plan_features

```

```

CREATE TABLE plan_features (
    plan_feature_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    plan_module_id UUID NOT NULL REFERENCES plan_modules(plan_module_id) ON DELETE
    RESTRICT,
    code VARCHAR(50) NOT NULL,
    name VARCHAR(200) NOT NULL,
    status SMALLINT NOT NULL DEFAULT \,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    created_by UUID,
    updated_at TIMESTAMPTZ,
    updated_by UUID,
    deleted_at TIMESTAMPTZ,
    deleted_by UUID,
    row_version BIGINT NOT NULL DEFAULT \
);

```

```

CREATE INDEX idx_plan_features_module ON plan_features(plan_module_id) WHERE
deleted_at IS NULL;

```

-- ۸. plan_version_features (pivot اصلاح نامگذاری بر اساس اصل ۱۱ و حذف کلمه ممنوعه)

```
CREATE TABLE plan_version_features (  
  plan_version_feature_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  plan_version_id UUID NOT NULL REFERENCES plan_versions(plan_version_id) ON DELETE  
  RESTRICT,  
  plan_feature_id UUID NOT NULL REFERENCES plan_features(plan_feature_id) ON DELETE  
  RESTRICT,  
  enabled BOOLEAN NOT NULL DEFAULT TRUE,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
  created_by UUID,  
  updated_at TIMESTAMPTZ,  
  updated_by UUID,  
  deleted_at TIMESTAMPTZ,  
  deleted_by UUID,  
  row_version BIGINT NOT NULL DEFAULT ۱  
);  
  
CREATE UNIQUE INDEX uq_plan_version_feature ON plan_version_features(plan_version_id,  
plan_feature_id) WHERE deleted_at IS NULL;
```

-- ۹. plan_offers

```
CREATE TABLE plan_offers (  
  plan_offer_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  plan_version_id UUID NOT NULL REFERENCES plan_versions(plan_version_id) ON DELETE  
  RESTRICT,  
  name VARCHAR(۲۰۰) NOT NULL,  
  status SMALLINT NOT NULL DEFAULT ۱,  
  start_date TIMESTAMPTZ,
```

```

end_date TIMESTAMPTZ,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT 1
);

```

```

CREATE INDEX idx_plan_offers_version ON plan_offers(plan_version_id) WHERE deleted_at IS
NULL;

```

```

-- 1. plan_offer_discounts

```

```

CREATE TABLE plan_offer_discounts (
    plan_offer_discount_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    plan_offer_id UUID NOT NULL REFERENCES plan_offers(plan_offer_id) ON DELETE RESTRICT,
    discount_value NUMERIC(10,2) NOT NULL,
    discount_type SMALLINT NOT NULL,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    created_by UUID,
    updated_at TIMESTAMPTZ,
    updated_by UUID,
    deleted_at TIMESTAMPTZ,
    deleted_by UUID,
    row_version BIGINT NOT NULL DEFAULT 1
);

```

```
CREATE INDEX idx_plan_offer_discounts_offer ON plan_offer_discounts(plan_offer_id) WHERE
deleted_at IS NULL;
```

```
-- 11. addons
```

```
CREATE TABLE addons (
  addon_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  code VARCHAR(50) NOT NULL,
  name VARCHAR(200) NOT NULL,
  status SMALLINT NOT NULL DEFAULT 1,
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  created_by UUID,
  updated_at TIMESTAMPTZ,
  updated_by UUID,
  deleted_at TIMESTAMPTZ,
  deleted_by UUID,
  row_version BIGINT NOT NULL DEFAULT 1
);
```

```
CREATE UNIQUE INDEX uq_addons_code ON addons(code) WHERE deleted_at IS NULL;
```

```
-- 12. offer_available_addons
```

```
CREATE TABLE offer_available_addons (
  offer_available_addon_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  plan_offer_id UUID NOT NULL REFERENCES plan_offers(plan_offer_id) ON DELETE RESTRICT,
  addon_id UUID NOT NULL REFERENCES addons(addon_id) ON DELETE RESTRICT,
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  created_by UUID,
  updated_at TIMESTAMPTZ,
```



```

updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT 1
);

CREATE UNIQUE INDEX uq_offer_addon ON offer_available_addons(plan_offer_id, addon_id)
WHERE deleted_at IS NULL;

-- =====

-- Layer 1: SaaS Business Layer (REVISED & SECURED) - PART 2

-- =====

-- 1.2. subscriptions

CREATE TABLE subscriptions (
subscription_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
tenant_id UUID NOT NULL REFERENCES tenants(tenant_id) ON DELETE RESTRICT,
plan_version_id UUID NOT NULL REFERENCES plan_versions(plan_version_id) ON DELETE
RESTRICT,
status SMALLINT NOT NULL DEFAULT 1,
start_date TIMESTAMPTZ,
end_date TIMESTAMPTZ,
next_billing_date TIMESTAMPTZ,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,

```

```
row_version BIGINT NOT NULL DEFAULT \
```

```
);
```

```
CREATE INDEX idx_subscriptions_tenant ON subscriptions(tenant_id) WHERE deleted_at IS  
NULL;
```

```
CREATE INDEX idx_subscriptions_version ON subscriptions(plan_version_id) WHERE deleted_at  
IS NULL;
```

```
-- \xi. subscription_events
```

```
CREATE TABLE subscription_events (
```

```
subscription_event_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```
subscription_id UUID NOT NULL REFERENCES subscriptions(subscription_id) ON DELETE  
RESTRICT,
```

```
event_type SMALLINT NOT NULL,
```

```
description VARCHAR(2000),
```

```
event_date TIMESTAMPTZ NOT NULL DEFAULT NOW(),
```

```
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
```

```
created_by UUID,
```

```
updated_at TIMESTAMPTZ,
```

```
updated_by UUID,
```

```
deleted_at TIMESTAMPTZ,
```

```
deleted_by UUID,
```

```
row_version BIGINT NOT NULL DEFAULT \
```

```
);
```

```
CREATE INDEX idx_subscription_events_sub ON subscription_events(subscription_id) WHERE  
deleted_at IS NULL;
```

```
-- \o. subscription_addons
```

```
CREATE TABLE subscription_addons (
```

```

subscription_addon_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
subscription_id UUID NOT NULL REFERENCES subscriptions(subscription_id) ON DELETE
RESTRICT,
addon_id UUID NOT NULL REFERENCES addons(addon_id) ON DELETE RESTRICT,
amount NUMERIC(۲۰,۴) NOT NULL,
status SMALLINT NOT NULL DEFAULT ۱,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT ۱
);

CREATE INDEX idx_subscription_addons_sub ON subscription_addons(subscription_id) WHERE
deleted_at IS NULL;

```

-- ۱۶. invoice_profiles

به لایه ۴ DDD اصلاح معماری: حذف فیلدهای آلوده هویتی و اتصال منطقی بر اساس اهداف --

```

CREATE TABLE invoice_profiles (
invoice_profile_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
tenant_id UUID NOT NULL REFERENCES tenants(tenant_id) ON DELETE RESTRICT,
billing_partner_id UUID NOT NULL, -- بدون bp_partners ارجاع منطقی به جدول
status SMALLINT NOT NULL DEFAULT ۱,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,

```

```

updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT 1
);

CREATE INDEX idx_invoice_profiles_tenant ON invoice_profiles(tenant_id) WHERE deleted_at IS
NULL;

```

-- 17. platform_invoices

```

CREATE TABLE platform_invoices (
    invoice_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    tenant_id UUID NOT NULL REFERENCES tenants(tenant_id) ON DELETE RESTRICT,
    invoice_profile_id UUID REFERENCES invoice_profiles(invoice_profile_id) ON DELETE SET NULL,
    invoice_number VARCHAR(100) NOT NULL,
    total_amount NUMERIC(10,2) NOT NULL,
    discount_amount NUMERIC(10,2) NOT NULL DEFAULT 0,
    tax_amount NUMERIC(10,2) NOT NULL DEFAULT 0,
    final_amount NUMERIC(10,2) NOT NULL,
    status SMALLINT NOT NULL DEFAULT 1,
    issue_date TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    due_date TIMESTAMPTZ,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    created_by UUID,
    updated_at TIMESTAMPTZ,
    updated_by UUID,
    deleted_at TIMESTAMPTZ,
    deleted_by UUID,

```

```
row_version BIGINT NOT NULL DEFAULT 1
```

```
);
```

```
CREATE UNIQUE INDEX uq_platform_invoices_number ON platform_invoices(invoice_number)  
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_platform_invoices_tenant ON platform_invoices(tenant_id) WHERE  
deleted_at IS NULL;
```

```
-- 18. platform_invoice_items
```

```
CREATE TABLE platform_invoice_items (
```

```
invoice_item_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```
invoice_id UUID NOT NULL REFERENCES platform_invoices(invoice_id) ON DELETE RESTRICT,
```

```
item_type VARCHAR(50) NOT NULL,
```

```
reference_id UUID,
```

```
description VARCHAR(500),
```

```
amount NUMERIC(10,2) NOT NULL,
```

```
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
```

```
created_by UUID,
```

```
updated_at TIMESTAMPTZ,
```

```
updated_by UUID,
```

```
deleted_at TIMESTAMPTZ,
```

```
deleted_by UUID,
```

```
row_version BIGINT NOT NULL DEFAULT 1
```

```
);
```

```
CREATE INDEX idx_platform_invoice_items_invoice ON platform_invoice_items(invoice_id)  
WHERE deleted_at IS NULL;
```

```
-- 19. platform_transactions
```

```

CREATE TABLE platform_transactions (
  transaction_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  invoice_id UUID NOT NULL REFERENCES platform_invoices(invoice_id) ON DELETE RESTRICT,
  gateway VARCHAR(100),
  transaction_number VARCHAR(200),
  amount NUMERIC(20,2) NOT NULL,
  status SMALLINT NOT NULL DEFAULT 1,
  transaction_date TIMESTAMPTZ,
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  created_by UUID,
  updated_at TIMESTAMPTZ,
  updated_by UUID,
  deleted_at TIMESTAMPTZ,
  deleted_by UUID,
  row_version BIGINT NOT NULL DEFAULT 1
);

CREATE INDEX idx_platform_transactions_invoice ON platform_transactions(invoice_id) WHERE
deleted_at IS NULL;

```

-- 2. tenant_wallet_transactions

```

CREATE TABLE tenant_wallet_transactions (
  wallet_transaction_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  wallet_id UUID NOT NULL REFERENCES tenant_wallets(wallet_id) ON DELETE RESTRICT,
  transaction_type SMALLINT NOT NULL,
  amount NUMERIC(20,2) NOT NULL,
  balance_after NUMERIC(20,2) NOT NULL,
  description VARCHAR(500),

```

```

created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT 1
);

```

```

CREATE INDEX idx_tenant_wallet_tx_wallet ON tenant_wallet_transactions(wallet_id) WHERE
deleted_at IS NULL;

```

-- 2. coupons

```

CREATE TABLE coupons (
coupon_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
code VARCHAR(100) NOT NULL,
discount_type SMALLINT NOT NULL,
discount_value NUMERIC(10,2) NOT NULL,
status SMALLINT NOT NULL DEFAULT 1,
start_date TIMESTAMPTZ,
end_date TIMESTAMPTZ,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT 1

```

```
);
```

```
CREATE UNIQUE INDEX uq_coupons_code ON coupons(code) WHERE deleted_at IS NULL;
```

```
-- ୧୧. coupon_usages
```

```
CREATE TABLE coupon_usages (
```

```
    coupon_usage_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```
    coupon_id UUID NOT NULL REFERENCES coupons(coupon_id) ON DELETE RESTRICT,
```

```
    tenant_id UUID NOT NULL REFERENCES tenants(tenant_id) ON DELETE RESTRICT,
```

```
    invoice_id UUID REFERENCES platform_invoices(invoice_id) ON DELETE SET NULL,
```

```
    discount_amount NUMERIC(୧୦,୧) NOT NULL,
```

```
    used_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
```

```
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
```

```
    created_by UUID,
```

```
    updated_at TIMESTAMPTZ,
```

```
    updated_by UUID,
```

```
    deleted_at TIMESTAMPTZ,
```

```
    deleted_by UUID,
```

```
    row_version BIGINT NOT NULL DEFAULT ୧
```

```
);
```

```
CREATE INDEX idx_coupon_usages_tenant ON coupon_usages(tenant_id) WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_coupon_usages_coupon ON coupon_usages(coupon_id) WHERE deleted_at IS NULL;
```

```
-- ୧୨. tenant_domains
```

```
CREATE TABLE tenant_domains (
```

```
    domain_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```



```
tenant_id UUID NOT NULL REFERENCES tenants(tenant_id) ON DELETE RESTRICT,  
domain_name VARCHAR(255) NOT NULL,  
domain_type SMALLINT NOT NULL DEFAULT 1,  
is_primary BOOLEAN NOT NULL DEFAULT FALSE,  
verification_token VARCHAR(255),  
verified_at TIMESTAMPTZ,  
ssl_status SMALLINT NOT NULL DEFAULT 0,  
status SMALLINT NOT NULL DEFAULT 1,  
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
created_by UUID,  
updated_at TIMESTAMPTZ,  
updated_by UUID,  
deleted_at TIMESTAMPTZ,  
deleted_by UUID,  
row_version BIGINT NOT NULL DEFAULT 1  
);
```

```
CREATE UNIQUE INDEX uq_tenant_domains_name ON tenant_domains(domain_name) WHERE  
deleted_at IS NULL;
```

```
CREATE INDEX idx_tenant_domains_tenant ON tenant_domains(tenant_id) WHERE deleted_at  
IS NULL;
```